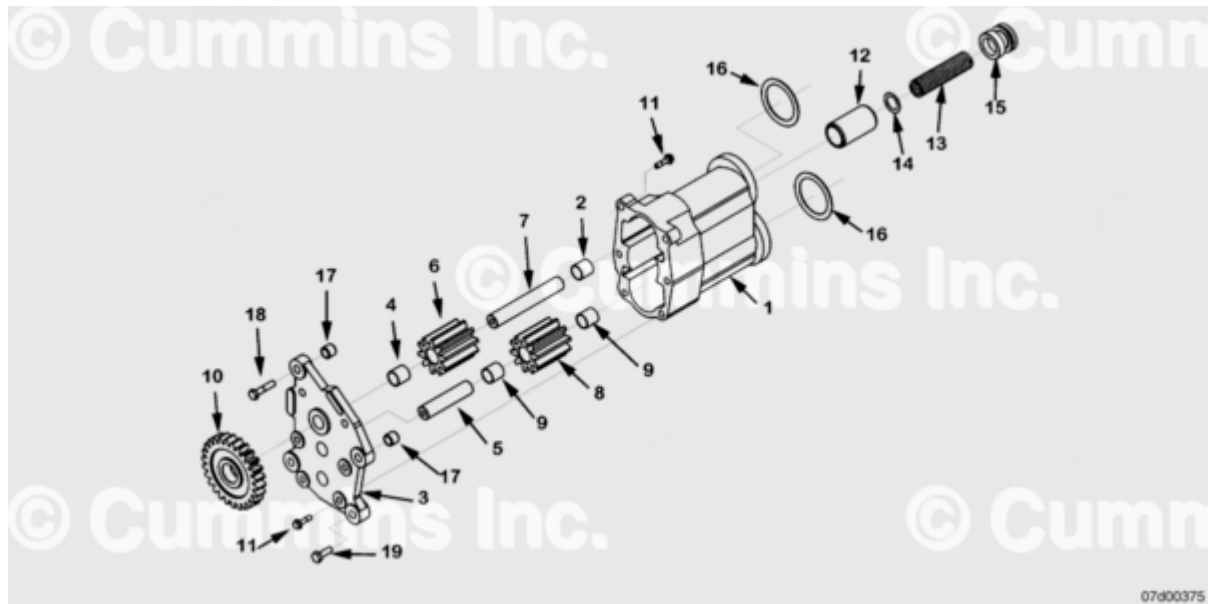


Trainings ▾

Exploded View



1. Oil pump body
2. Bushing
3. Oil pump cover
4. Bushing
5. Oil pump shaft
6. Oil pump gear
7. Oil pump shaft
8. Oil pump gear
9. Bushing
10. Oil pump gear
11. Flange head capscrew
12. Bypass valve
13. Compression spring
14. Shim
15. Spring retainer
16. O-ring seal
17. Capscrew bushing
18. Flange head capscrew
19. Capscrew.

Preparatory Steps

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery last.

- Disconnect the batteries or air starter to prevent accidental starting.
- Drain the lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>) in Section 7.
- Remove the oil pan. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>) in Section 7.
- Remove the block stiffener plate. Refer to Procedure 001-089 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>) in Section 1.
- Remove the fan and fan spacer(s). Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/89/89-008-040.html>) in Section 8.
- Remove the fan belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/89/89-008-002.html>) in Section 8.
- Remove the alternator belt. Refer to Procedure 013-005 (</qs3/pubsys2/xml/en/procedures/89/89-013-005.html>) in Section 13.
- Remove the alternator. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/89/89-013-001.html>) in Section 13.
- Remove the fan hub. Refer to Procedure 008-036 (</qs3/pubsys2/xml/en/procedures/89/89-008-036-tr.html>) in Section 8.
- Remove the fan belt tensioner. Refer to Procedure 008-036 (</qs3/pubsys2/xml/en/procedures/89/89-008-036-tr.html>) in Section 8.
- Remove the vibration damper. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/89/89-001-052.html>) in Section 1.
- Remove the front crankshaft seal. Refer to Procedure 001-023 (</qs3/pubsys2/xml/en/procedures/89/89-001-023.html>) in Section 1.
- Remove the alternator drive gear and shaft. Refer to Procedure 009-009 (</qs3/pubsys2/xml/en/procedures/89/89-009-009.html>) in Section 9.
- Remove the front gear cover. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/89/89-001-031.html>) in Section 1.

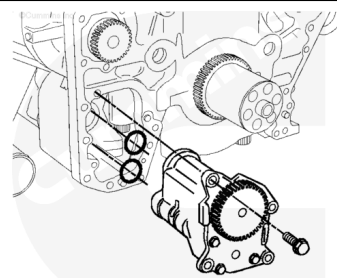
- Remove the oil pump idler gear. Refer to Procedure 001-098 in Section 1 (/qs3/pubsys2/xml/en/procedures/89/89-001-098.html).

Remove

Remove the four capscrews from the oil pump mounting flange.

Use a pry bar to gently pry the oil pump out of the engine block.

Remove and discard the two o-rings and the seal ring from the pump.

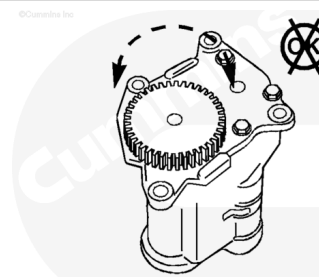


07400240

Clean and Inspect for Reuse

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



07400241

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

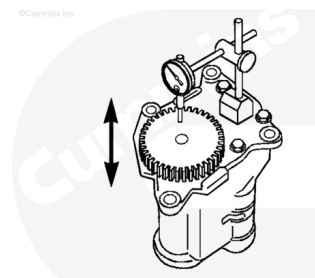
Clean and inspect the oil pump for reuse. Rotate the drive gear slowly and check the pump gears for damage.

Measure the oil pump end clearance.

Oil Pump End Clearance		
mm		in
0.04	MIN	0.002

Oil Pump End Clearance

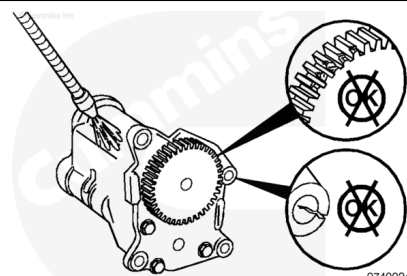
mm		in
0.097	MAX	0.004



07400242

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



07400243

Use solvent to flush the lubricating oil pump.

Inspect the gears of the oil pump for damage. If the part is damaged, it **must** be rebuilt or replaced.

Disassemble

Remove the four capscrews.

Tap on the dowel with a mallet to separate the cover from the oil pump body.

Remove the oil pump cover, idler gear, and shaft from body.

Support the oil pump body on an arbor press.

Press the shaft through the drive gear.

⚠ WARNING ⚠

To reduce the possibility of personal injury, hold the regulator stop in position when removing the roll pin. The regulator stop is under pressure and can pop out.

Use an arbor press and mandrel to hold the retainer in place while the dowel pin is being removed.

Some models use a short dowel and roll pin or a snap ring to hold the retainer.

Remove the dowel pin with a brass drift.

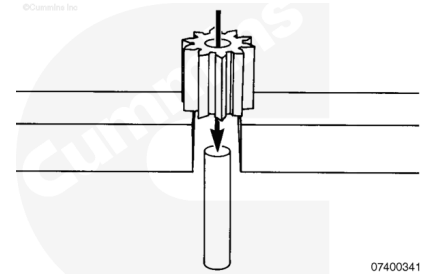
Slowly raise the arbor press to relieve the spring tension.

Remove the retainer, spring, and the bypass valve.

Note : Only disassemble the gear and shaft when the gear or shaft must be replaced.

Support the gear and shaft in an arbor press.

Press the shaft through the gear.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the oil pump parts with solvent.

Dry with compressed air.

Check the free length of the spring.

Spring Free Length (Approximately)

mm	in
113.6	4.5

Measure the spring force (7) at the working height (6) with valve spring tester, Part Number 3375182.

Spring Working Height

mm	in
76.5	3.01

Spring Force

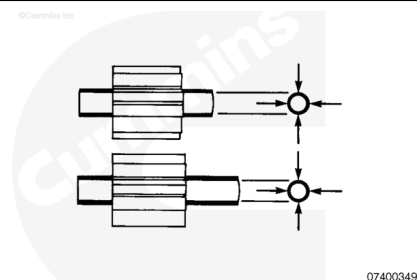
N		lbf
824	MIN	185
911	MAX	205

If the spring is **not** within specifications, it **must** be replaced.

Inspect the gears for damage.

If the gears are damaged they **must** be replaced.

Measure the oil pump shaft outside diameter.



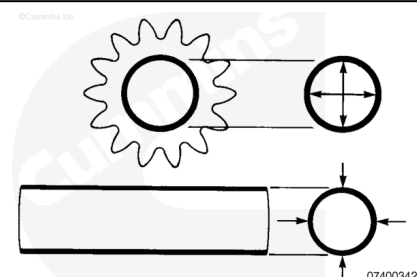
Oil Pump Shaft Diameter

mm		in
18.09	MIN	0.712
18.11	MAX	0.713

If the oil pump shaft is **not** within specifications, the oil pump shaft **must** be replaced.

Note : Do not disassemble the gear and shaft to make this measurement. Only measure the gear inside diameter if the shaft is replaced.

Measure the oil pump gear inside diameter.



Oil Pump Gear Inside Diameter

mm		in
18.028	MIN	0.710
18.049	MAX	0.711

Inspect the oil pump housing and cover for damage.

If the oil pump body or cover are damaged, the damaged component **must** be replaced.

Install

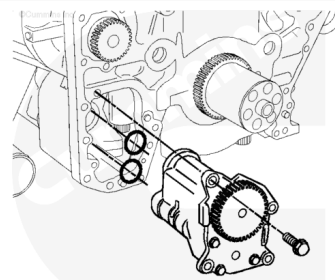
Apply a light coating of grease to the seal rings to hold them in place during assembly.

Install the seal rings on the pump.

Use vegetable oil to lubricate the seal rings and the bore in the block.

Install the pump and the capscrews. Tighten the capscrews.

Torque Value: 66 n•m [50 ft-lb]

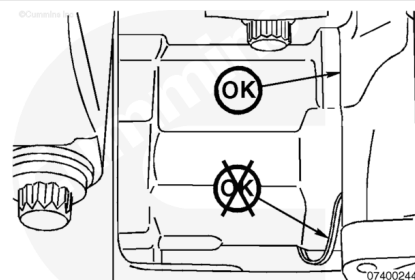


07400240

Be sure the seal rings at the back of the pump are positioned correctly.

Look up into the block between the number 1 and number 2 main bearing caps. The seal **must not** protrude from the space between the rear of the pump and the block.

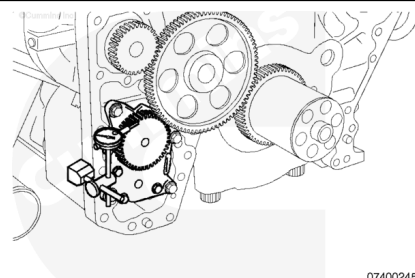
Install the oil pump idler gear. Refer to Procedure 001-098 (</qs3/pubsys2/xml/en/procedures/89/89-001-098.html>).



07400244

Check the oil pump gear backlash. Refer to Procedure 001-055 (</qs3/pubsys2/xml/en/procedures/89/89-001-055.html>).

Adjust the backlash, if necessary, by loosening the mounting capscrews and rotating the pump.



07400245

Lubricating Oil Pump Gear Backlash

mm		in
0.121	MIN	0.004
0.333	MAX	0.013

Finishing Steps

- Install the front gear cover. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/89/89-001-031.html>) in Section 1.
- Install the front crankshaft seal. Refer to Procedure 001-023 (</qs3/pubsys2/xml/en/procedures/89/89-001-023.html>) in Section 1.
- Install the alternator drive gear and shaft. Refer to Procedure 009-009 (</qs3/pubsys2/xml/en/procedures/89/89-009-009.html>) in Section 9.

- Install the vibration damper. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/89/89-001-052.html>) in Section 1.
- Install the fan belt tensioner. Refer to Procedure 008-036 (</qs3/pubsys2/xml/en/procedures/89/89-008-036-tr.html>) in Section 8.
- Install the fan hub. Refer to Procedure 008-036 (</qs3/pubsys2/xml/en/procedures/89/89-008-036-tr.html>) in Section 8.
- Install the alternator. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/89/89-013-001.html>) in Section 13.
- Install the alternator belt. Refer to Procedure 013-005 (</qs3/pubsys2/xml/en/procedures/89/89-013-005.html>) Section 13.
- Install the fan belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/89/89-008-002.html>) in Section 8.
- Install the fan and fan spacer(s). Refer to Procedure 008-040 (</qs3/pubsys2/xml/en/procedures/89/89-008-040.html>) in Section 8.
- Install the block stiffener plate. Refer to Procedure 001-089 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>) in Section 1.
- Install the oil pan. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>) in Section 7.
- Fill the engine with clean, 15W-40 oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>) in Section 7.
- Connect the batteries or air starter.
- Operate the engine. Check for leaks.